Day-3 Amplitudes: `B02_compton`

Generated at `2026-08-12T22:06:31.565733+08:00` from AmplitudeIR `amplitude_ir:b02_compton:generate`

Momentum Definitions

$$q_s = p_1 + k_1$$

$$q_u = p_1 - k_2$$

Diagram Amplitudes

$$\mathcal{M}_s = \bar{u}(p_2)_{a_2}\,(-ie\gamma^{\mu_4})\,\epsilon^*_{\mu_4}(k_2)\frac{i(\not{q}_s + m_e)}{q_s^2 - m_e^2 + i\epsilon}\,(-ie\gamma^{\mu_3})\,\epsilon_{\mu_3}(k_1)u(p_1)_{a_1} \tag{1}$$

$$\mathcal{M}_u = \bar{u}(p_2)_{a_2}\,(-ie\gamma^{\mu_4})\,\epsilon_{\mu_4}(k_1)\frac{i(\not{q}_u + m_e)}{q_u^2 - m_e^2 + i\epsilon}\,(-ie\gamma^{\mu_3})\,\epsilon^*_{\mu_3}(k_2)u(p_1)_{a_1} \tag{2}$$

Total Amplitude

$$\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u \tag{3}$$

Rule/Source Audit

amplitude	rule id	source
`amplitude:b02_compton:amp_s`	`rule:qed_tree_v1:e_e_gamma_vertex`	`docs/QED_CONVENTION_AUDIT.md`
`amplitude:b02_compton:amp_s`	`rule:qed_tree_v1:electron_propagator`	`docs/QED_CONVENTION_AUDIT.md`
`amplitude:b02_compton:amp_u`	`rule:qed_tree_v1:e_e_gamma_vertex`	`docs/QED_CONVENTION_AUDIT.md`
`amplitude:b02_compton:amp_u`	`rule:qed_tree_v1:electron_propagator`	`docs/QED_CONVENTION_AUDIT.md`